



YOLO-FaceV2: A scale and occlusion aware face detector

Ziping Yu^a, Hongbo Huang^{b,*}, Weijun Chen^c, Yongxin Su^d, Yahui Liu^e, Xiuying Wang^b^a School of Instrument Science and Opto-electronic Engineering, Beijing Information Science and Technology University, Beijing, China^b Computer School, Beijing Information Science and Technology University, Beijing, China^c Data Algorithm NIO, Shanghai, China^d School of Mechanical and Electrical Engineering, Beijing Information Science and Technology University, Beijing, China^e School of Information Management, Beijing Information Science and Technology University, Beijing, China

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ABSTRACT

In recent years, face detection algorithms based on deep learning have made great progress. Nevertheless, the effective utilization of face detectors for small and occlusion faces remains challenging, primarily stemming from the limitations in pixel information and the presence of missing features. In this paper, we propose a novel real-time face detector, YOLO-FaceV2, built upon the YOLOv5 architecture. Our approach introduces a Receptive Field Enhancement (RFE) module designed to extract multi-scale pixel information and augment the receptive field for accurately detecting small faces. To address issues related to face occlusion, we introduce an attention mechanism termed the Separated and Enhancement Attention Module (SEAM), which effectively focuses on the regions affected by occlusion. Furthermore, we propose a Slide Weight Function (SWF) to mitigate the imbalance between easy and hard samples. The experiments demonstrate that our YOLO-FaceV2 achieves performance exceeding the state-of-the-art on the WiderFace validation dataset. Source code and pre-trained model are available at <https://github.com/Krasjet-Yu/YOLO-FaceV2>.

1. Introduction

Face detection plays a pivotal role in various applications related to facial analysis, such as face recognition, verification, and attribute analysis. With the booming of deep convolutional neural networks in recent years, the performance of face detectors has been greatly improved. Numerous high-performance face detection algorithms leveraging deep learning have been introduced. Generally, these algorithms can be categorized into two main branches. One branch of typical deep-learning-based face detection algorithms [1] uses cascading means of neural networks as feature extractors and classifiers to detect faces from coarse to fine. Despite their great success, it is noteworthy that cascade detectors are plagued by certain limitations, such as challenges in training and slow detection speeds. The other branch stems from enhancements made to general-purpose object detection algorithms. These detectors are designed to capture more generalized object features, allowing task-specific detectors to leverage this broader information and further refine performance through specialized designs. Notable examples of such face detectors include YOLO [2], Faster R-CNN [3] and RetinaNet [4] fall into this category. In this study, inspired

by YOLOv5 [2], we propose a novel face detector named YOLO-FaceV2, which achieves state-of-the-art performance in one-stage face detection.

Although deep convolutional networks have improved face detection remarkably, detecting faces with high variance in scale in real-world scenes remains a substantial challenge. On the one hand, the scale of face changes continuously, and the existing algorithms are usually based on discrete anchors, so it is difficult to adapt to this continuous variation, thereby hindering the detection of faces at different scales. To address this issue, researchers have proposed several solutions. In our previous work, we proposed the YOLO-Face [5], an enhanced face detector based on YOLOv3, focusing primarily on scale variance. This involved designing anchor ratios tailored for human faces and employing a more precise regression loss function. Zhang et al. devised the size of the anchors based on the effective receptive field in S^3FD [6]. FaceBoxes [7] designed multi-scale anchors to enrich the receptive fields and discretizes anchors over different layers to handle faces of various scales. Thus, inspired by the design of receptive fields, we design a novel Receptive Field Enhancement Module (RFE) to provide a more diverse receptive field, facilitating improved capture of faces at various scales. Additionally, we integrated an effective

* Corresponding author.

E-mail address: hbb@bistu.edu.cn (H. Huang).<https://doi.org/10.1016/j.patcog.2024.110714>

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receptive field approach [8] to design our multi-scale anchor, further enhancing the suitability of our detector for faces of varying sizes.

On the other hand, in certain scenarios, such as surveillance environments, faces may appear extremely small. Omitting the detection of such targets may lead to failures of the intelligent surveillance systems. Detecting tiny faces, e.g., those less than 16×16 pixels, necessitates tiling numerous small anchors across the image, improving the recall rate but exacerbating the problem of extreme class imbalance, a major contributor to excessive false positives [9]. During training, the cumulative impact of these false positives predominates the model update process, resulting in inefficient training due to imbalanced samples. To address this issue, Lin et al. introduced Focal Loss to dynamically assign more weights to the difficult sample examples [4]. Similarly, the Gradient Harmonizing Mechanism (GHM) [10] suppressed gradients of simpler samples to prioritize difficult ones, while Prime Sample Attention (PISA) [11] assigned weights to positive and negative samples based on different criteria. However, existing hard sample mining methods often require numerous hyperparameters, posing practical inconveniences. Hence, we propose a Slide Weight Function named SWF to adjust the weight of imbalanced samples, enabling adaptive learning of positive and negative sample threshold parameters. Additionally, we employ the Normalized Wasserstein Distance metric [12] to alleviate the impact of slight IoU offset on small-scale face detection.

Face occlusion frequently arises in densely populated scenes, often resulting in detector failures to accurately detect faces, thereby reducing system accuracy and recall. Occlusion of the face leads to partial data loss, making it challenging for the model to extract sufficient features for precise localization and detection. To tackle this challenge, some researchers have explored the use of attention mechanisms for facial feature extraction [13]. FAN [14] proposed anchor-level attention, emphasizing the maintenance of response values in unobstructed regions while compensating for reduced response values in obscured regions through attention mechanisms. However, this approach fails to fully exploit inter-channel information. Motivated by this, we introduce a novel attention module named Separated and Enhancement Attention Module (SEAM) to focus on missing region and enhance the representation information of occluded faces. On the other hand, the overlap of faces in crowded scenes makes detectors sensitive to non-maximum suppression (NMS) thresholds, leading to missed detections. Thus, we modify the Repulsion Loss, originally designed for handling overlapping pedestrians, to tailor face detection and improve the recall of intra-class occluded faces.

In this paper, we aim to tackle the challenges associated with scale variations and occlusion in face detection by devising a novel face detector based on YOLOv5. Our objective is to develop an optimal combinatorial detector capable of effectively addressing issues such as large-scale variations, occluded faces, and imbalanced samples caused by small faces. To this end, we present a novel face detector named YOLO-FaceV2, with the following key contributions:

- We introduce a RFE module to facilitate more effective learning of different receptive fields within the feature map, thereby enhancing the representation ability of the feature pyramid. Additionally, inspired by the effective receptive field, we design a multi-scale anchor and integrate Normalized Wasserstein Distance (NWD) to mitigate the dependency of small-scale faces on Intersection over Union (IoU).
- We propose an attention module SEAM to improve representation learning for occluded faces. Moreover, we employ Repulsion Loss for face detection, penalizing the shift of predicted boxes towards other ground-truth objects and ensuring a certain distance between predicted boxes of different targets to reduce sensitivity to NMS.
- We devise a weighted function called SWF to adaptively assign higher weights to hard samples, facilitating the model to learn more challenging features.

2. Related works

Face Detection. Face detection has been a prominent area in computer vision for decades. During the early stages of deep learning, face detection algorithms usually relied on neural networks for automatic feature extraction and classification. For instance, CascadeCNN [15] proposed a cascaded structure comprising three stages of carefully designed deep convolutional networks, which predicted face and landmark locations in a coarse-to-fine manner. Similarly, MTCNN [1] introduced a cascade architecture to jointly align face landmarks and detect face locations. PCN [16] utilized an angle prediction network to rectify the face angle and enhance face detection accuracy. However, early deep learning-based face detection methods suffered from drawbacks such as laborious training, susceptibility to local optima, slow detection speeds, and low accuracy.

Recent advancements in face detection algorithms have largely been influenced by the strengths of generic object detection techniques, including SSD [17], Faster R-CNN [3], and RetinaNet [4]. Zhang et al. [7] developed a lightweight network, FaceBoxes, based on the SSD structure. It rapidly reduced feature size by 32x through down-sampling and employed a multi-scale network module to enrich features in both width and depth dimensions. SRN [18] an enhancement of the RefineDet [19] and RetinaNet [4] object detection algorithms, achieved high performance by introducing two-stage classification and regression and incorporating a multi-branch module to amplify the effects of receptive fields. Additionally, MTCNet [20] leveraged facial landmarks to enhance the detection performance through collaboration between face detection and face alignment.

Scale-invariance. Large variations in face scale within complex scenes pose significant challenges to face detection algorithms, impacting their accuracy. The multi-scale detection capability mainly depends on the scale-invariance features, and many works address this problem to extract features more accurately and effectively [21]. For detecting small objects, using fewer down-sampling layers and dilated convolution can significantly improve the detection performance [22]. Another approach to tackle this challenge involves utilizing more anchors. Anchors provide valuable prior information, and employing denser anchors along with corresponding matching strategies can effectively improve the quality of object proposals [6]. Multi-scale training can also be beneficial for constructing image pyramids and increasing sample diversity, offering a simple yet effective means to enhance multi-scale object detection performance. However, as receptive fields expand and semantic information becomes richer, spatial information may correspondingly be lost. One natural approach is to integrate deep semantic information with shallow features [23]. Additionally, TridentNet [24] and SNIP [25] also proposed novel solutions to address the multi-scale problem, which will be elaborated upon in subsequent sections.

Occlusion problem. Crowding faces and the following occlusion problem give rise to partial data and lack of information about the occluded faces, because some regions are invisible or the boundaries are blurred, which can easily cause missed detection and low recalls. Prior research has highlighted the importance of contextual information in mitigating the occlusion problem in face detection. SSH [26] incorporated context by utilizing simple convolution layers to expand the window around candidate proposals. FAN [14] introduced anchor-level attention to detect occluded faces by emphasizing features from the face region. PyramidBox [27] implemented a context-sensitive prediction module, replacing the convolution layers of the context module in SSH with the residual prediction module of DSSD. Additionally, RetinaFace [28] applied independent context modules on five feature pyramid levels to increase the receptive field and enhance the rigid context modeling power. These approaches have demonstrated effectiveness in addressing the occlusion problem. Therefore, leveraging contextual information to improve the effectiveness of occluded regions represents a viable direction worthy of further exploration.

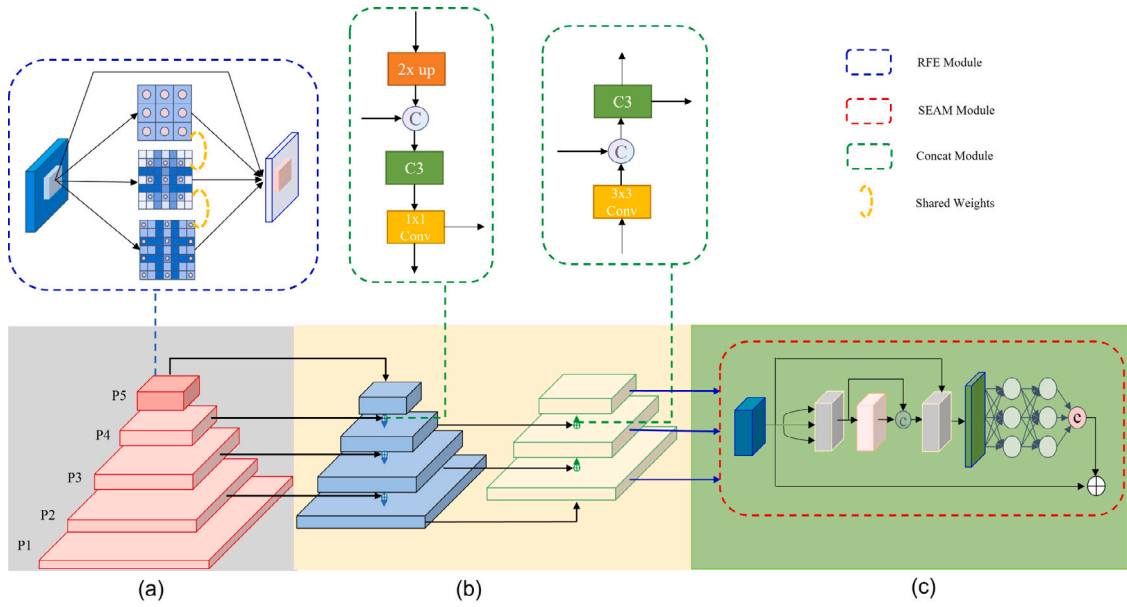


Fig. 1. Network architecture of YOLO-FaceV2. (a) Backbone: a feed-forward CSPDarknet53 architecture extracts the multi-scale feature maps. To expand the receptive field, the CSP block in P5 is replaced by the receptive field enhancement module (RFE) which is shown in the blue dotted box. (b) Neck: a Spatial Pyramid Pooling (SPP) separates out the most significant context features and increases the receptive field and a Path Aggregation Network (PAN) aggregates parameter from different backbone levels for different detector levels. To compensate for increased receptive field loss of resolution, the P2 layer is fused into PAN which is shown between (a) and (b). (c) a Separated and Enhancement Attention Module (SEAM) uses the relationship between feature maps to recall occluded features which is shown in the red dotted box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Imbalance of easy and hard samples. In the realm of one-stage face detection, a substantial proportion of samples are categorized as easy, leading to their dominance in total losses. Consequently, the model tends to prioritize learning features from easy samples while neglecting those from hard ones. To mitigate this issue, the OHEM [29] algorithm selected difficult samples based on sample loss and incorporates their loss into stochastic gradient descent training. However, the OHEM algorithm may overlook easy samples, prompting the development of Focal Loss [4], which assigns weights to all samples to enhance their contribution, thereby yielding higher accuracy. This idea is also followed by SRN [18]. Faceboxes [7] sorted the samples according to their IoU loss, and controls the ratio of positive to negative samples to be less than 1:3. While these methods alleviate sample imbalance to some extent, they introduce additional hyperparameters, thereby complicating model training. Therefore, we propose a sample balance function with adaptive parameters to achieve more effective sample balancing.

3. YOLO-FaceV2

3.1. Network architecture

YOLOv5 is an excellent general object detector. We introduce YOLOv5 into the face detection field and try to solve the problems of multi-scale faces and face occlusion, etc.

The architecture of our YOLO-FaceV2 detector, depicted in Fig. 1. It consists of three main components: the backbone, the neck and the heads. For the backbone, we employ CSPDarknet53 and integrate the RFE module into the P5 layer to facilitate the fusion of multi-scale features. Retaining the PAN structure from PAN [30], the neck section incorporates the P2 layer to enhance the detection accuracy of small faces. Additionally, we introduce a dedicated module within the heads to bolster the model's occlusion detection capabilities.

In Fig. 1(a), the left portion illustrates the backbone, consisting of CSP and CBS blocks tasked with feature extraction from input images. Notably, the RFE module in the P5 layer is instrumental in expanding the effective receptive field and enhancing multi-scale fusion

capabilities. In Fig. 1(b), the neck layers, depicted in blue and yellow, encompass SPP and PAN modules. Lastly, in Fig. 1(c), the introduction of the SEAM module enhances the responsiveness to occluded faces following the neck layer.

3.2. Scale invariance

Scale-Aware RFE Model. Since different size of receptive fields means different capability of capturing long-range dependency, we design the RFE module to sufficiently make use of the advantage of receptive fields in the feature map by using dilated convolution. Inspired by TridentNet, we use four branches of different rates of dilated convolution to capture multi-scale information and different ranges of dependency. All the branches have sharing weights, the only difference is their distinctive receptive fields. On the one hand, it reduces the amount of parameters and thus the risk of potential overfitting. On the other hand, it can make full use of each sample. The proposed RFE module can be divided into two parts: multi branches based on dilated convolutions and the gathering & weighting layer, as shown in Fig. 2. The multi-branch part takes 1, 2 and 3 as the rates of different dilated convolutions separately, which all use fixed convolutional kernel size 3×3 . Furthermore, we add a residual connection to prevent the problem of gradient explosion and disappearance during training. The gathering and weighting layer is used to gather information from different branches and weight every branch of the features. The weighting operation is used to balance the representation of different branches.

To make it clear, we replace the bottleneck of C3 module in YOLOv5 with RFE module to increase the receptive field of feature map, so as to improve the accuracy of multi-scale target detection and recognition, as shown in Fig. 2.

Anchor Design Strategy. Anchor design strategy is critical in face detection. In our model, each of the three detection heads is associated with specific anchor scales. The anchor design encompasses both the width-to-height ratio and anchor size, as outlined in Table 1. To determine the width-to-height ratio, we derive statistics from the WiderFace training set based on ground-truth face ratios. In our face detection task, we set the aspect ratio to 1:1.2 based on these statistics. Regarding

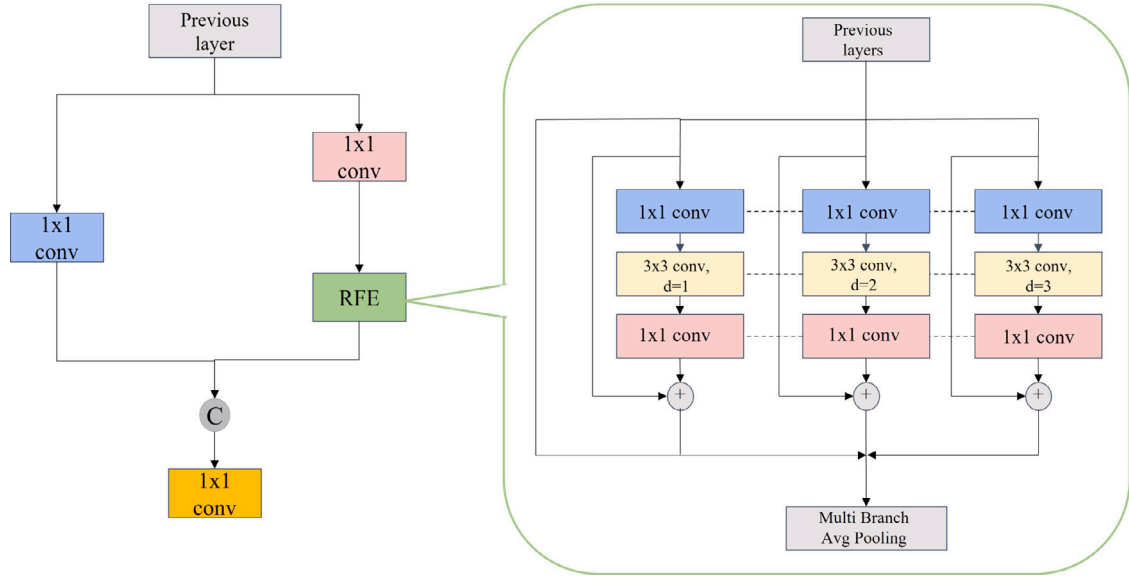


Fig. 2. Modified CSP block and RFE module. For CSP block in P5, we replace bottleneck with RFE. The right figure shows the detailed architecture of the RFE. It consists of 1×1 convolution, 3×3 convolution with different dilated rate and average pooling layer.

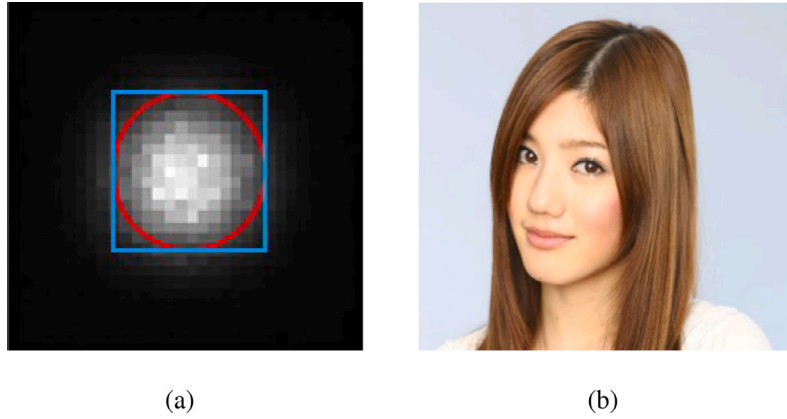


Fig. 3. (a) **Effective receptive field:** The whole black box is the theoretical receptive field (TRF) and the white circle with Gaussian distribution is the effective receptive field (ERF). The figure is from [8]. (b) **A special example:** The whole box is the origin anchor box setting by TRF and the blue box is the new anchor estimated by the red circle which is the ERF. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1

The three detection layers and nine anchor scales.

Layer	Stride	Ratio	Anchor
P2	4	1.2	[16, 20.16, 25.40]
P3	8	1.2	[32, 40.32, 50.80]
P4	16	1.2	[64, 80.63, 101.59]

anchor sizes, we initially compute the theoretical receptive field of P2 based on the distribution of convolutional and pooling layers. However, not all pixels within the theoretical receptive field contribute equally to the final output. In essence, only a fraction of the area significantly influences the output value, as depicted in Fig. 3(a). According to [8], the effective receptive field exhibits near Gaussian shapes and has $1/\sqrt{n}$ relative shrinkage (as illustrated in Fig. 3(b)). Finally, we determine the anchor sizes of P3 and P4, maintaining fixed stride sizes with P2 as detailed in Table 1.

Normalized Gaussian Wasserstein Distance. Normalized Wasserstein Distance is a new similarity measurement used in small target detection. Firstly, the bounding box is modeled as a two-dimensional

Gaussian distribution, and the similarity between the predicted target and the real target is calculated through their corresponding Gaussian distribution, that is, the normalized Wasserstein distance between them is calculated according to Eq. (1). For the detected targets, whether they overlap or not, they can be measured by the distribution similarity. The NWD is not sensitive to the scale of the targets and thus is more suitable for measuring the similarity between small targets. In our regression loss function, we add a NWD Loss to make up for the disadvantage of the IoU loss for small target detection. But we still keep the IoU loss because it is suitable for large object detection.

$$NWD(\mathcal{N}_a, \mathcal{N}_b) = \exp \left(- \frac{\sqrt{W_2^2(\mathcal{N}_a, \mathcal{N}_b)}}{C} \right) \quad (1)$$

$$W_2^2(\mathcal{N}_a, \mathcal{N}_b) = \left\| \left(\left[cx_a, cy_a, \frac{w_a}{2}, \frac{h_a}{2} \right]^T, \left[cx_b, cy_b, \frac{w_b}{2}, \frac{h_b}{2} \right]^T \right) \right\|_2^2 \quad (2)$$

Where C is a constant closely related to the dataset, $W_2^2(\mathcal{N}_a, \mathcal{N}_b)$ is a distance measure, and $\mathcal{N}_a$ and $\mathcal{N}_b$ are Gaussian distributions modeled by $A = (cx_a, cy_a, w_a, h_a)$ and $B = (cx_b, cy_b, w_b, h_b)$.

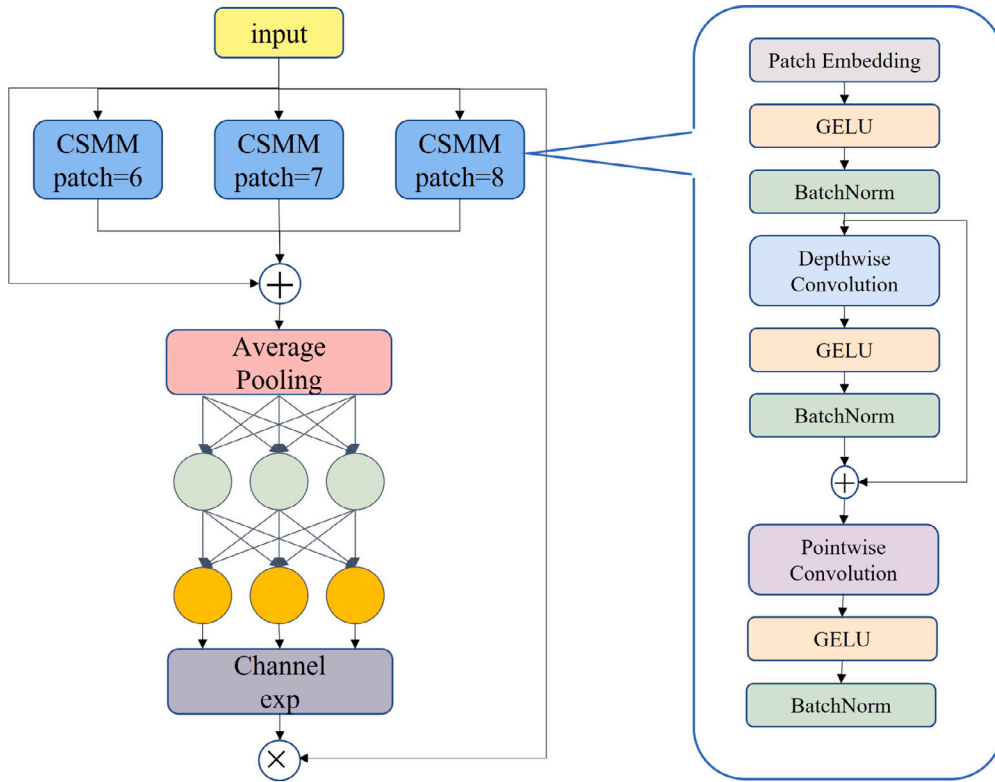


Fig. 4. Illustration of SEAM. The left is the architecture of the SEAM, the right part is the structure of CSMM(channel and spatial mixing module). The CSMM utilizes different patch to multi-scale feature and uses depth separable convolution to learn the correlation of spatial dimensions and channels.

3.3. Occlusion aware network

Occlusion-Aware Attention Network. We integrate the multi-head attention network, denoted as the SEAM module (refer to Fig. 4), to address inter-class occlusion, manifesting as alignment errors, local aliasing, and feature loss. The SEAM module serves three primary objectives: facilitating multi-scale face detection, highlighting facial regions within images, and weakening the background area oppositely. The first part of SEAM is the depthwise separable convolution with residual connection. Depthwise separable convolution is operated depth-by-depth, that is, the convolution separated channel-by-channel. Although depthwise separable convolution can learn the importance of different channels and reduce the amount of parameters, it ignores the information relationship between channels. To make up for this loss, the outputs of different depthwise convolutions are subsequently combined by point-by-point (1×1) convolutions. Subsequently, a two-layer full connection network aggregates information from each channel, enhancing inter-channel connectivity. This framework aims to compensate for occlusion-induced information loss by leveraging learned relationships between obstructed and unobstructed facial regions. The logits obtained from the fully connected layer undergo exponential transformation to expand the value range from $[0, 1]$ to $[1, e]$. This exponential normalization provides a monotonic mapping relationship that makes the results more tolerant to position error. Finally, the SEAM module's output, treated as attention weights, is multiplied with the original features, enabling the model to effectively address facial occlusion.

Occlusion-Aware Repulsion Loss. Intra-class occlusion may cause face A contains the features of face B, resulting in a higher false detection rate. The introduction of the repulsion loss can effectively alleviate this problem through repulsion. The repulsion loss is divided into two parts: RepGT and RepBox. The function of RepGT Loss is to make the current bounding box as far away from the surrounding

ground-truth box as possible. The surrounding ground-truth box here refers to the face label with the largest IoU with the face except for the object to be returned by the bounding box itself. The formula of RepGT loss function is as follows:

$$L_{\text{RepGT}} = \frac{\sum_{P \in P_+} \text{Smooth}_{\text{In}} \left(\text{IoG} \left(P, G_{\text{Rep}}^P \right) \right)}{|P_+|} \quad (3)$$

where

$$\text{Smooth}_{\text{In}} = \begin{cases} -\ln(1-x) & x \leq \sigma \\ \frac{x-\sigma}{1-\sigma} - \ln(1-\sigma) & x > \sigma \end{cases} \quad (4)$$

P in the formula is the face prediction frame, G_{Rep}^P is the ground-truth with the largest IoU around the face. The overlap between P and G_{Rep}^P is defined as intersection over ground-truth (IoG): $\text{IoG}(P, G) = \frac{\text{area}(P \cap G)}{\text{area}(G)}$ and $\text{IoG}(B, G) \in [0, 1]$. $\text{Smooth}_{\text{In}}$ is $(0, 1)$ a continuously differentiable. In function, $\sigma \in [0, 1]$ is a smoothing parameter to adjust the sensitivity of repulsion loss to outliers.

The purpose of RepBox loss is to make the prediction frame as far away from the surrounding prediction frame as possible and reduce the IoU between them, so as to avoid one of the prediction frames belonging to two faces being suppressed by NMS. We divide the prediction frame into multiple groups. Assuming there are individual faces, the division form is shown in Eq. (5). The prediction frames between the same groups return to the same face label, and the prediction frames between different groups correspond to different face labels.

$$\rho_+ = \rho_1 \cap \rho_2 \cap \dots \cap \rho_{|g|} \quad (5)$$

Then, for the prediction box between different groups p_i and p_j , we hope that we get the smaller the overlap area between p_i and p_j . RepBox also uses $\text{Smooth}_{\text{In}}$ as an optimization function. The overall loss function is as follows:

$$L_{\text{RepBox}} = \frac{\sum_{i \neq j} \text{Smooth}_{\text{In}} \left(\text{IoU} \left(B^{p_i}, B^{p_j} \right) \right)}{\sum_{i \neq j} 1 [\text{IoU} \left(B^{p_i}, B^{p_j} \right) > 0] + \epsilon} \quad (6)$$

Table 2

Results of the ablation experiments for scale-invariance verification on the Widerface validation dataset.

Method	Detail	Easy	Medium	Hard	Params(M)	Flops(G)
baseline	-	94.65	93.00	83.30	7.063	16.4
RFE Module	w/o P2	93.67	92.14	83.87	6.101	16.0
	w/ P2	95.06	93.60	85.47	5.097	16.0
NWD Loss	w/o IoU	91.13	84.4	75.77	7.063	16.4
	w/ IoU	94.82	93.87	84.31	7.063	16.4
Anchor Design	w/ theoretical	94.65	93.00	83.30	7.063	16.4
	w/ effective	94.89	93.75	84.20	7.063	16.4

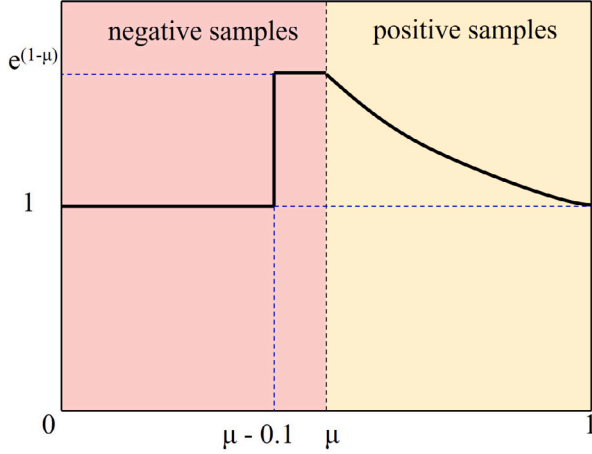


Fig. 5. We propose a novel loss we term the Slide Loss that adaptively learns the positive and negative sample threshold parameters μ . Setting high weights near μ increases the relative loss for hard-classified examples, putting more focus on hard, misclassified examples.

3.4. Sample weighting function

The sample imbalance problem, i.e., in most cases the number of easy samples are quite large while hard samples are relatively sparse, has attracted the attention of many researchers. In our work, we design a SWF which looks like a “slide” to address this problem. The distinction between easy and hard samples is based on the IoU size of the prediction box and ground-truth box. To reduce the hyperparameters, we take the average of the IoU value of all bounding boxes as the threshold value μ , those less than μ are taken as negative samples, and positive samples for the values greater than μ . However, samples near the boundaries often suffer from large losses due to unclear classifications. We hope that the model can learn to optimize these samples and use these samples more sufficiently to train the network. Nevertheless, the number of such samples is relatively small. Therefore, we try to assign higher weights to the difficult samples. We first divide the samples into positive and negative samples by the parameter μ . Then, we emphasize on the samples at the boundary by a weighting function Slide, as shown in Fig. 5. The Slide weighting function can be expressed as Eq. (7).

$$f(x) = \begin{cases} 1 & x \leq \mu - 0.1 \\ e^{1-\mu} & \mu < x < \mu - 0.1 \\ e^{1-x} & x \geq \mu \end{cases} \quad (7)$$

4. Experiments

In this part, we provide the details for experiments and results to demonstrate the performance and efficiency of our face detector.

4.1. Dataset

According to the detection difficulties, the dataset can be divided into three parts: easy, medium and hard, which roughly correspond to large, medium and small faces, respectively. Hence it is suitable to be used as the evaluation benchmark of our model. We trained our model on the WiderFace training set and evaluated it on the validation set. We generally believe that if the recall rate on the hard set reaches over 90%, it can be said that the detector has achieved quite good performance.

The Fddb dataset contains 2845 images, with a total of 5171 human faces. For its large variety of difficult postural angles, out-of-focus faces and extensive occlusion, it is used here to test the effectiveness of the occlusion-aware modules. And we test them on the Fddb dataset without retraining on them and use the mAP as the performance metric. When the mAP on the Fddb dataset reaches 95% or above, we consider the occlusion problem to be effectively addressed.

4.2. Implementation details

We use YOLOv5 as our baseline and the methods are implemented by PyTorch. The hyperparameters of data enhancement, training and post-processing are the same as those of YOLOv5, i.e. mosaic probability, learning rate and IoU threshold for NMS. The gain of RepGT α and RepBox β is set to 0.01 and 0.1, respectively. The sample threshold μ for SWF is 0.5. We pretrain the complete model on the ImageNet dataset. Then the face task is fine-tuned about 50 iterations with a batch size of 16 images on 3090Ti.

According to the problem we need to solve, we divided the experiment into three parts: scale invariance, occlusion problem and sample imbalance. To solve the face scale problem, we present the effects of the scale-aware RFE model, NWD loss and anchor design strategy, incrementally. To verify the solution of face occlusion, we introduce the Fddb dataset containing a large number of obscured faces to demonstrate SEAM module and repulsion loss performance and use the mAP as the performance metric. We conduct SWF to solve sample imbalance. Finally, we compare the performances between our proposed detector and other SOTA face detectors on the WiderFace dataset to show the robustness of face detector.

4.3. Scale invariance study

In this section, we conduct comprehensive experiments of scale invariance module on the WiderFace dataset to evaluation their affections on the performance of the model. Then the modules are analyzed one-by-one.

For RFE Module: The performance of RFE module is shown in the Table 2. According to the second row of Table 2, it can be observed that the hard subset has increased by 0.57% when only RFE module is used. Then, we fuse P2 layer features to the PAN structure, so that the fused feature map contains more information of small targets. The effect is shown in the third row of Table 2. The mAP has increased on the easy, medium and hard subset by 0.5%, 0.6% and 2.17%, respectively. The significant improvement on the hard subset demonstrates the effectiveness of the RFE module.

Table 3

Results of the ablation experiments for occlusion-aware verification on the WiderFace validation dataset.

Method	Network	Easy	Medium	Hard
Baseline	YOLOv5	94.65	93.00	83.30
+SE [31]		94.47	92.90	82.74
+CBAM [31]		95.31	93.80	84.03
+EMA [32]		95.02	93.03	83.77
+SEAM (ours)		95.53	93.82	84.36
Repulsion Loss		95.27	93.63	83.80

Table 4

Results of the mAP metric on the Fddb dataset.

Method	mAP
PyramidBox [27]	0.9869
FaceBoxes [7]	0.9598
YOLO5Face [33]	0.9843
YOLO-FaceV2 (ours)	0.9871

Table 5

Ablation study results on the WiderFace validation dataset.

SWF	RFE	SEAM	Easy	Medium	Hard
✗	✗	✗	96.30	94.71	86.11
✓	✗	✗	96.85	95.67	86.57
✓	✓	✗	97.79	96.34	87.26
✓	✓	✓	98.39	97.07	89.39

For NWD Loss: The mAP of NWD loss and combined loss of NWD and IoU are listed in the fourth and fifth lines of the Table 2. We have first adopted the NWD Loss instead of IoU as the regression loss. However, the mAP fell sharply. The possible reason is that for large and medium-sized targets, the effect measured by IoU is better than NWD, and NWD can effectively improve the detection accuracy of small targets. Therefore, we choose to preserve the IoU Loss and improve the robustness of the model for small target detection by adjusting the ratio between them to 1:1. From the table, we see that the mAP improves by 0.17%, 0.87% and 1.01% on the easy, medium and hard subset, respectively.

For Anchor Design: According to the receptive field and face shape characteristics, the performance impact of designed anchors is shown in the seventh row of Table 2. It improves by 0.24%, 0.75%, 0.9% on the easy, medium and hard datasets respectively. As we would expect, properly designed anchors help to recall more small face targets.

Fig. 7(a), (c) and (e) are some actual detection examples about small faces on WiderFace by YOLO-FaceV2.

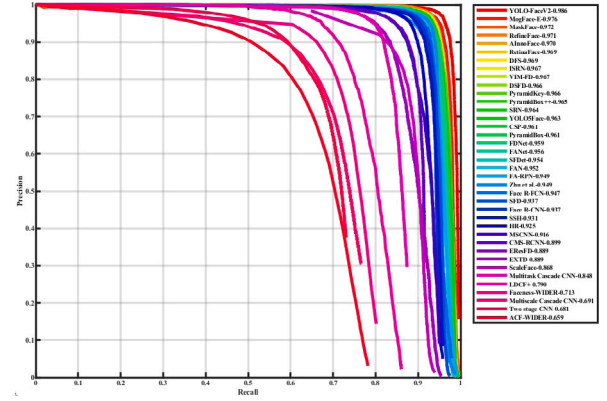
4.4. Occlusion aware study

In this section, we conduct comprehensive study to analyze the effectiveness of occlusion-aware methods. And the modules are evaluated on the WiderFace and Fddb dataset.

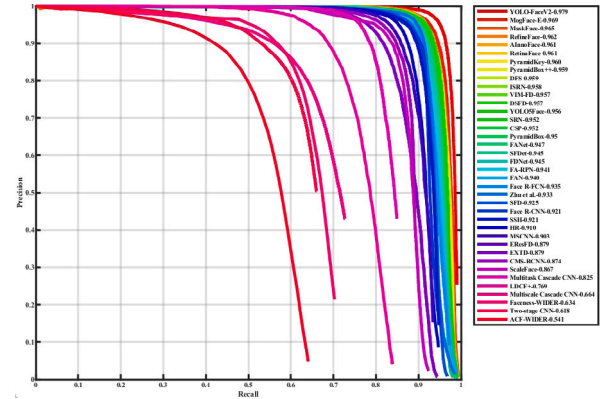
For SEAM Module: Our proposed SEAM block is the attention network. The results are shown in the Table 3. We have conducted comparative experiments with several mainstream attention modules, including SE [31], CBAM [31], and EMA [32]. As we can see, SEAM model outperforms all other attention-based models and the mAP increases by 0.88%, 0.82% and 1.06% on the WiderFace compared to the baseline. This shows the superiority of our attention module.

For Balance of RepGT and RepBox: According to the results in the sixth row of the Table 3, the repulsion loss function improves the model accuracy by 0.71%, 0.63% and 0.5% on easy, medium and hard subsets.

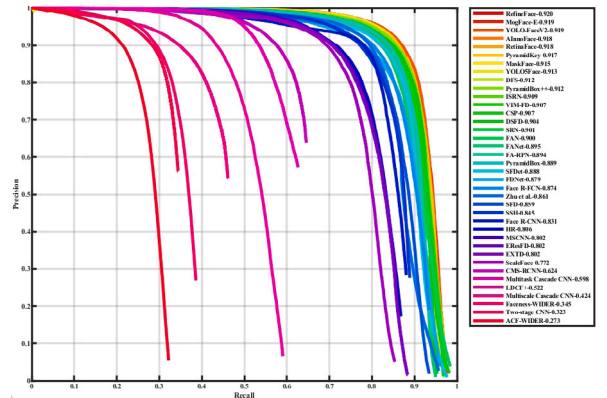
For Evaluation on Fddb: In order to demonstrate the occlusion-aware capability of our model, we test it on the Fddb dataset. The Fddb dataset has a large number of obscured faces, which can fully evaluate the effectiveness of the proposed method. The results are listed



(a) Easy



(b) Medium



(c) Hard

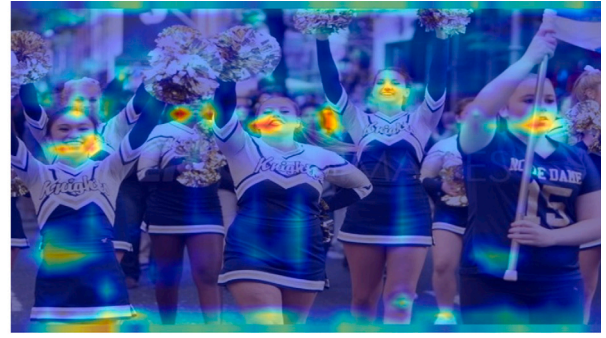
Fig. 6. Detection results on the WiderFace validation dataset. (a) results on the Easy dataset (b) results on the Medium dataset (c) results on the Hard dataset.

in Table 4. Our model reaches 98.71% mAP and achieves state-of-the-art performance in Fddb.

Fig. 7(b), (d) and (f) illustrate the visualization of the heatmap of the occluded face via Gradient-based Localization when using only the baseline, repulsion loss and SEAM module, respectively. (b) represents the result of the baseline network. The face blocked by the arm on the right of the picture is not noticed. (d) shows the feature map that only repulsion loss is added. The face on the right obscured by the arm is



(a)



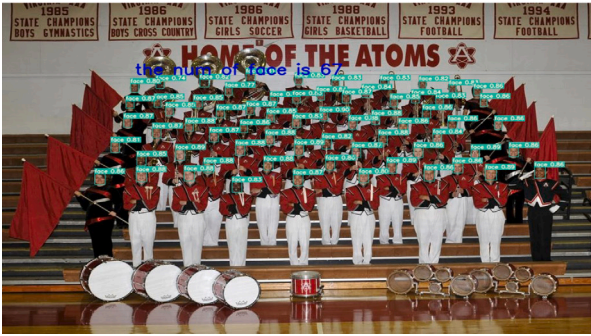
(b)



(c)



(d)



(e)



(f)

Fig. 7. (a), (c) and (e) are some examples of detected small faces using YOLO-FaceV2. (b), (d) and (f) are the visualization of feature maps produced by models with occlusion-aware methods.

detected and the faces of the two near each other on the left are also well distinguished. (f) is the result of implementing the SEAM module. As we can see, each face is well focused, and all obscured faces are recalled. The visualized results well support the effect of our model on occlusion-aware.

4.5. Sample imbalance

The main purpose of the SWF is to make the model focus more on hard samples. According to the results in the first row of the Table 5, the SWF slightly improves the accuracy of the model on the medium and hard subsets.

4.6. Effect of model ensemble

In addition, tackling all issues at once makes it difficult to assess the novelty. Therefore, we mainly study the effectiveness of the main

module combinations, and take other methods borrowed from the literature as the baseline, as shown in the first row of Table 5. The sequence of module combinations is shown in Table 5. And it shows that our three innovative modules have all resulted in significant metric improvements on the WiderFace dataset.

4.7. Comparisons with existing face detectors

We mainly compare with various excellent face detectors presented recently as shown in Table 7. The results of the methods are classified according to the original general detectors such as Fast R-CNN, SSD, YOLO, etc. Compared with face detection algorithms based on SSD and R-CNN, our YOLO-FaceV2 achieves the state-of-the-art performance on easy, medium and hard subset, relatively. In the YOLO series, we look at the performance of relatively large models whose number of parameters is larger than 3M and the number of flops is larger than 5G.

Table 6

A series of YOLO models with different parameters.

Model	Easy	Medium	Hard
YOLO-Facev2n	0.936	0.921	0.812
YOLO-FaceV2s	0.983	0.970	0.893
YOLO-FaceV2 m	0.984	0.973	0.905
YOLO-FaceV2l	0.986	0.979	0.919

Table 7

Comparison of our YOLO-FaceV2 and existing face detectors on the WiderFace validation dataset.

Method	Detector	Easy	Medium	Hard
Faster R-CNN				
	HR	0.925	0.91	0.806
	Face R-CNN	0.937	0.921	0.831
	FDNet	0.959	0.945	0.879
SSD				
	SFD	0.937	0.925	0.859
	SSH	0.931	0.921	0.845
	PyramidBox	0.961	0.95	0.889
	SFDet	0.954	0.945	0.888
RetinaNet				
	FAN	0.952	0.94	0.9
	SRN	0.964	0.952	0.901
	RetinaFace	0.969	0.961	0.918
	RefineFace	0.971	0.962	0.92
YOLO				
	YOLO-FaceV1	0.899	0.872	0.693
	YOLOv5l-Face	0.963	0.954	0.908
	YOLOv7	0.969	0.955	0.880
	YOLO-FaceV2l	0.986	0.979	0.919

Our YOLO-FaceV2l achieves 98.6%, 97.9%, 91.9% on the three subsets, exceeding the previous SOTA by 2.3%, 2.5%, 1.1%. This also indicates that we have further released the model performance within the YOLO framework.

However, our detector performed a little bit worse compared to RetinaFace on the hard subset. The reason is that the RetinaNet-based face detectors adopted two-stage classification and regression, which makes the results appear higher in accuracy. But it is achieved with more computational resource consumption, and our method is more efficient. In addition, we implement a series of models with different number of parameters, as listed in Table 6. We implement four relatively large models, include nano, small, medium and large size. Experiments show that with the increase of model parameters, AP still shows an increasing trend. Using a larger number of parameters may lead to better results.

And the precision–recall (PR) curves of our YOLO-FaceV2 face detector, along with the competitors, are shown in Fig. 6. Our face detector achieves exceeding SOTA performance on the small and medium subset, and the result of hard subset perform better than the YOLO series methods.

5. Conclusion

In this work, we emphasize the significance of addressing facial scale variability and occlusion occurrences, as these factors significantly contribute to missed detections by the detector. It indicates that solving these problems during object detection training is critical for improving the recall rate. We analyze the reasons behind this problem, and devise a RFE module aimed at augmenting the receptive field to effectively accommodate faces of varying scales. Additionally, we introduce the SWF to mitigate the imbalance between easy and hard samples induced by small faces. Moreover, we propose the SEAM to guide the model in focusing on occluded facial features. Experimental results demonstrate the efficacy of our three main contributions, culminating

in YOLO-FaceV2 achieving performance comparable to or surpassing the state-of-the-art on the WiderFace validation dataset. Furthermore, it is worth mentioning that although we have proposed models with different number of parameters, it appears that these models have not yet reached their full potential. Further exploration of larger model capacities remains as part of our future work, aiming to continuously improve the accuracy and efficiency of face detection under complex conditions.

CRedit authorship contribution statement

Ziping Yu: Methodology, Software, Visualization, Writing – original draft. **Hongbo Huang:** Methodology, Writing – original draft, Writing – review & editing. **Weijun Chen:** Methodology. **Yongxin Su:** Visualization. **Yahui Liu:** Writing – review & editing. **Xiuying Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Hongbo Huang is an associate professor at the Computer School of Beijing Information Science and Technology University, China. He received his Ph.D. degree in Control Science and Engineering in 2015 from University of Science and Technology Beijing. His research interests include computer vision, deep learning and artificial intelligence.



Yahui Liu is an associate professor at the School of Information Management in Beijing Information Science and Technology University, China. She received her Ph.D. degree in Mechanical and Electronic Engineering from Beijing University of Posts and Telecommunications in 2011. Her research interests include Computer Vision and Data Analysis.



Xiuying Wang is a lecturer at the Computer School of Beijing Information Science and Technology University, China. She received her Ph.D. degree from Chinese Academy of Sciences University in 2019. Her research interests include image processing, natural language processing, Human-machine Conversation System.